

HAMADA, Japan

WMO: 477550

Lat: 34.8967N

Lon: 132.0706E

Elev: 68

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.2	33.9	18.5	14.0	37.7	20.6	15.5	38.6	31.5	45.3	27.4	42.3	10.9	60	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	89.8	78.3	87.8	78.2	86.2	77.5	80.3	86.4	79.2	85.3	78.3	84.4	8.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.8	84.3	77.4	143.1	83.4	76.5	138.4	82.6	44.0	86.5	42.9	85.5	41.9	84.6	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.1	21.2	18.7		DB	28.3	94.4	2.8	2.7	26.3	96.3	24.6	97.9	23.0	99.4	21.0	101.3
				WB	25.6	81.8	2.5	1.3	23.8	82.7	22.3	83.5	20.9	84.2	19.1	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	42.8	43.4	48.0	56.1	64.1	70.6	78.4	80.7	73.4	64.3	55.7	47.3
	DBStd	13.73	4.36	5.01	5.54	5.60	4.40	4.00	4.08	3.50	4.62	4.53	4.97	4.96
	HDD50	656	226	194	106	11	0	0	0	0	0	0	8	111
	HDD65	3057	687	604	527	271	69	3	0	0	1	66	281	548
	CDD50	4495	4	10	44	196	438	618	881	953	701	444	178	28
	CDD65	1421	0	0	0	5	42	171	416	488	252	45	2	0
	CDH74	11240	0	0	2	20	138	616	3726	5084	1524	128	2	0
	CDH80	3314	0	0	0	1	10	78	1098	1809	304	14	0	0
	WSAvg	8.2	10.3	9.5	9.3	8.6	7.3	6.4	7.3	6.8	7.0	7.5	8.6	10.1
	PrecAvg	64.9	4.0	3.1	4.8	4.3	6.3	6.9	10.0	5.2	7.2	4.2	4.6	3.9
(15) Precipitation	PrecMax	93.5	6.0	5.4	7.8	8.6	11.1	15.5	25.1	17.1	13.0	12.1	7.2	7.8
	PrecMin	44.4	1.3	1.5	2.2	1.3	3.1	0.7	0.4	0.9	3.3	0.9	1.3	0.8
	PrecStd	13.1	1.3	1.2	1.8	2.1	2.2	4.3	6.9	3.9	3.3	2.5	1.9	1.9
	PrecStd	13.1	1.3	1.2	1.8	2.1	2.2	4.3	6.9	3.9	3.3	2.5	1.9	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	60.2	65.6	69.6	76.8	81.3	85.2	91.4	94.4	88.9	81.7	73.0	63.5	(19)
	0.4% MCWB	54.2	56.1	57.4	63.2	66.3	74.3	77.9	77.4	77.3	72.4	63.3	56.1	(20)
	2% DB	55.4	59.5	64.6	71.9	77.5	82.3	88.7	91.0	85.7	77.2	69.4	60.3	(21)
	2% MCWB	48.7	53.1	54.4	60.3	65.2	73.2	78.2	78.0	76.6	68.7	61.5	53.1	(22)
	5% DB	52.8	55.3	61.0	68.8	74.9	79.6	86.8	88.9	83.5	74.6	67.1	57.9	(23)
	5% MCWB	46.2	49.1	53.0	58.5	63.5	71.2	77.6	78.4	75.4	66.5	59.6	50.9	(24)
	10% DB	50.5	52.2	58.1	66.1	72.7	77.4	85.3	87.3	81.3	72.5	64.8	55.6	(25)
	10% MCWB	44.2	46.1	51.4	57.3	62.6	70.1	76.9	78.2	74.1	64.9	57.7	49.1	(26)
	0.4% WB	55.0	58.1	60.0	66.9	70.9	76.5	81.5	81.9	79.4	73.7	66.0	58.8	(27)
	0.4% MCDB	58.7	62.9	65.2	73.0	76.3	81.8	87.0	87.5	85.0	79.4	70.1	62.3	(28)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	50.5	54.1	56.9	63.0	68.2	74.9	79.9	80.8	77.8	71.2	63.5	54.8	(29)
	2% MCDB	54.2	58.7	61.9	68.6	73.4	79.8	85.7	86.9	83.5	75.7	67.5	58.6	(30)
	5% WB	47.0	50.2	54.7	60.4	66.5	73.6	78.8	79.8	76.5	68.8	61.2	52.2	(31)
	5% MCDB	51.5	54.1	59.5	65.8	71.7	77.8	84.7	86.1	82.0	72.9	65.7	56.5	(32)
	10% WB	44.8	47.1	52.3	58.8	64.9	72.1	77.8	78.9	75.2	66.4	58.8	49.8	(33)
	10% MCDB	49.6	51.1	57.3	64.1	70.2	75.8	83.7	85.3	79.9	70.7	63.7	54.6	(34)
(35) Mean Daily Temperature Range	MDBR	10.2	12.1	14.0	14.9	14.3	11.5	10.1	11.3	12.1	13.7	13.0	10.8	(35)
	5% DB MCDBR	14.7	17.6	18.4	18.5	17.8	14.0	12.6	13.9	13.1	15.6	15.8	14.8	(36)
	5% DB MCWBR	11.0	13.3	13.0	11.7	10.4	7.5	6.2	6.6	7.3	9.1	10.6	10.6	(37)
	5% WB MCDBR	13.8	15.8	16.4	15.1	14.0	10.5	11.3	11.7	11.8	12.9	13.8	13.3	(38)
	5% WB MCWBR	11.4	13.0	12.9	10.6	9.5	5.9	6.4	6.7	6.9	8.4	10.4	11.0	(39)
	5% WB MCWBR	11.4	13.0	12.9	10.6	9.5	5.9	6.4	6.7	6.9	8.4	10.4	11.0	(39)
(40) Clear-Sky Solar Irradiance	taub	0.439	0.481	0.552	0.576	0.572	0.591	0.588	0.547	0.503	0.460	0.440	0.430	(40)
	taud	2.006	1.932	1.785	1.781	1.843	1.879	1.939	2.058	2.121	2.128	2.090	2.043	(41)
	Ebn at Noon	229	235	229	233	235	231	230	237	241	240	228	223	(42)
	Edh at Noon	43	52	66	69	66	63	59	52	47	43	40	40	(43)
(44) All-Sky Solar Radiation	RadAvg	625	859	1212	1528	1699	1536	1568	1630	1285	1072	761	567	(44)
	RadStd	49	79	100	128	176	136	226	169	116	102	62	37	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page